MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 12** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 30**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

MATHS MANTHAN | $(5 \times 1 = 5$ Marks)

Q1. The angle of elevation of the top of a cliff 100 m high from a point directly at its base is: [1]

(A) $0°$ (B) $45°$ (C) $90°$ (D) undefined

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Justification: Looking straight up from the base along the vertical cliff line of sight forms an angle of $\boxed{90°}$ with the horizontal ground.

Q2. If the angle of elevation of a cloud from a point h metres above a lake is θ, then the angle of depression of its reflection in the lake is: [1]

(A) equal to θ (B) greater than θ (C) less than θ (D) $90° - \theta$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Justification: With cloud at height $H > h$ above water and horizontal offset x, $\tan\theta = \dfrac{H-h}{x} < \dfrac{H+h}{x} = \tan\phi \implies \phi$ is $\boxed{\text{greater than } \theta}$.

Q3. The shadow of a tower is 30 m long when the sun's elevation is $30°$. The length of the shadow when the sun's elevation is $60°$ is: [1]

(A) 10 m (B) 15 m (C) $10\sqrt{3}$ m (D) 20 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Tower height $h = 30\tan 30° = \dfrac{30}{\sqrt{3}} = 10\sqrt{3}$ m. Shadow at $60°$ is $s = \dfrac{h}{\tan 60°} = \dfrac{10\sqrt{3}}{\sqrt{3}} = \boxed{10 \text{ m}}$.

Q4. A man on the deck of a ship, 10 m above water level, observes that the angle of elevation of the top of a cliff is 60° and the angle of depression of the base is 30°. The distance of the cliff from the ship is: [1]

(A) $10\sqrt{3}$ m (B) $20\sqrt{3}$ m (C) 30 m (D) 10 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: From the 10 m deck, base depression is 30° $\implies$ horizontal distance $= 10\cot 30° = \boxed{10\sqrt{3}\text{ m}}$.

Q5. Assertion (A): In a right-angled triangle formed by a vertical pole, its shadow, and the sun's rays, the hypotenuse represents the line of sight from the tip of the shadow to the sun.
Reason (R): The line of sight is defined as the line drawn from the observer's eye to the point being observed. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: The sun's ray grazing the top of the pole to the shadow tip coincides with the observer's line of sight from shadow tip to the sun. Both (A) and (R) are true and (R) explains (A).

SECTION B: Very Short Answer Questions

MATHS MANTHAN | $(2 \times 2 = 4$ Marks)

Q6. From a point P on the ground, the angle of elevation of the top of a 10 m tall building is 30°. A flag is hoisted at the top of the building and the angle of elevation of the top of the flagstaff from P is 45°. Find the length of the flagstaff. [Take $\sqrt{3} = 1.732$] [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

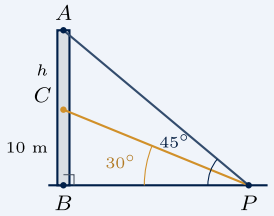
Let building be $BC = 10$ m and flagstaff be $AC = h$ m, observed from P ($\angle CPB = 30°, \angle APB = 45°$).

[0.5 Mark]

In $\triangle CBP$: $BP = 10\sqrt{3}$ m. In $\triangle ABP$: $10 + h = BP\tan 45° = 10\sqrt{3}$. [1 Mark]

$$\implies h = 10(\sqrt{3} - 1) = 10(1.732 - 1) = \boxed{7.32\text{ m}}$$

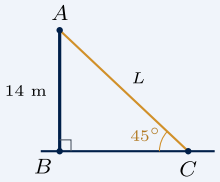
Length of flagstaff: [0.5 Mark]



Q7. A steel wire is attached to the top of an electric pole of height 14 m to anchor it firmly to the ground. If the wire is inclined at 45° to the horizontal, find the length of the wire. [Take $\sqrt{2} = 1.414$] **[2]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

Let pole $AB = 14$ m and wire AC be inclined at $\angle ACB = 45°$. [0.5 Mark]



$$\sin 45° = \frac{14}{AC} \implies AC = 14\sqrt{2}$$

$$\implies AC = 14 \times 1.414 = \boxed{14\sqrt{2} \text{ m} \approx 19.796 \text{ m}}$$

Final length of wire: [1.5 Marks]

SECTION C: Short Answer Questions

MATHS MANTHAN | $(2 \times 3 = 6$ Marks)

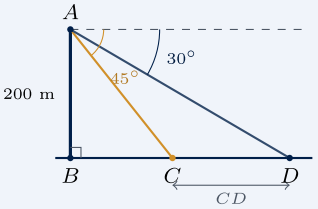
Q8. An observer on the top of a cliff 200 m high observes two cargo ships moving away from the cliff in the sea in the same direction. If the angles of depression of the ships change from 45° to 30°, find the distance between the two ships. [Take $\sqrt{3} = 1.732$] **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let cliff $AB = 200$ m and ship positions on same side be C (45°) and D (30°). [0.5 Mark]

In $\triangle ABC$: $BC = 200 \cot 45° = 200$ m. [1 Mark]

In $\triangle ABD$: $BD = 200 \cot 30° = 200\sqrt{3}$ m. [1 Mark]



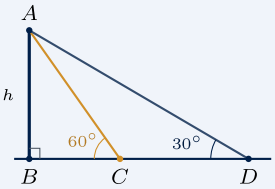
Distance between the two positions/ships $= CD = 200\sqrt{3} - 200 = 200(1.732 - 1) = 200(0.732) =$ $\boxed{146.4 \text{ m}}$. [0.5 Mark]

Q9. A car is approaching a tower at a uniform speed. The angle of elevation of the top of the tower from the car changes from 30° to 60° in 12 minutes. How much more time will the car take to reach the base of the tower? **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let tower $AB = h$, car start at D ($\angle ADB = 30°$) and reach C ($\angle ACB = 60°$) in 12 min. [0.5 Mark]

$BC = \frac{h}{\sqrt{3}}$, $BD = h\sqrt{3} \implies DC = h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}} = 2\,BC.$ [1.5 Marks]



Since uniform speed covers $DC = 2\,BC$ in 12 minutes, remaining half-distance CB takes:

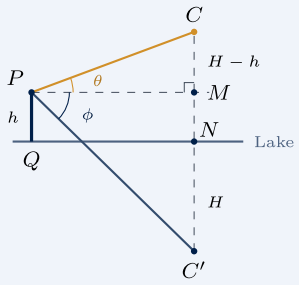
$$\text{More time required} = \frac{12}{2} = \boxed{6 \text{ minutes}}$$

Final remaining time: [1 Mark]

Q10. The angle of elevation of a stationary cloud from a point h metres above the surface of a lake is θ and the angle of depression of its reflection in the lake is ϕ. Prove that the height of the cloud above the lake surface is $H = h\left(\dfrac{\tan\phi + \tan\theta}{\tan\phi - \tan\theta}\right)$. If $h = 250$ m, $\theta = 30°$, and $\phi = 60°$, calculate the numerical height of the cloud above the lake. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let observer P be h above lake QN, cloud C at height H above lake ($CM = H - h$), reflection C' at depth H below lake ($C'M = H + h$), and $PM = x$. [0.5 Mark]



In $\triangle PMC$: $x = \dfrac{H-h}{\tan\theta}$; in $\triangle PMC'$: $x = \dfrac{H+h}{\tan\phi}$. Equating:

$$(H-h)\tan\phi = (H+h)\tan\theta \implies \boxed{H = h\left(\frac{\tan\phi + \tan\theta}{\tan\phi - \tan\theta}\right)}$$

[Proved]. [2.5 Marks]

For $h = 250$ m, $\theta = 30°, \phi = 60°$: $\tan 60° = \sqrt{3}$, $\tan 30° = \dfrac{1}{\sqrt{3}}$.

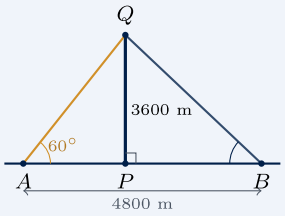
$$H = 250\left(\frac{\sqrt{3} + 1/\sqrt{3}}{\sqrt{3} - 1/\sqrt{3}}\right) = 250\left(\frac{4/\sqrt{3}}{2/\sqrt{3}}\right) = 250 \times 2 = \boxed{500 \text{ m}}$$

Numerical evaluation: [2 Marks]

Q11. An aeroplane flying at a constant altitude of 3600 m is observed simultaneously from two ground radar stations A and B situated 4800 m apart on level ground. If the aeroplane is between the two stations in the same vertical plane and the angle of elevation from station A is $60°$, calculate the distance of the aeroplane from station B. [Take $\sqrt{3} = 1.732$] **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let aeroplane Q hover at height $QP = 3600$ m above point P between stations A and B ($AB = 4800$ m), with $\angle QAP = 60°$. [1 Mark]



In right $\triangle QPA$: $\tan 60° = \dfrac{3600}{AP} \implies AP = \dfrac{3600}{\sqrt{3}} = 1200\sqrt{3} = 1200(1.732) = 2078.4$ m. [1.5 Marks]

Horizontal ground distance from P to station B: $PB = AB - AP = 4800 - 2078.4 = 2721.6$ m. [1 Mark]

Direct slant distance QB of the aeroplane from station B (by Pythagoras theorem in right $\triangle QPB$):

$$QB = \sqrt{QP^2 + PB^2} = \sqrt{3600^2 + 2721.6^2} = \sqrt{12{,}960{,}000 + 7{,}407{,}106.56}$$

$$= \sqrt{20{,}367{,}106.56} \approx \boxed{4512.99 \text{ m} \approx 4513 \text{ m}}$$

(*Note: Horizontal distance $PB = 2721.6$ m; direct line-of-sight distance $QB \approx 4513$ m.*) [1.5 Marks]

Q12. CASE STUDY: Tactical Drone Aerial Reconnaissance and Border Security

A surveillance drone D hovers at constant altitude $DP = 150$ m directly above checkpoint P. It spots an approaching vehicle at V_1 (depression $30°$); 3 minutes later, the vehicle reaches V_2 (depression $60°$).

(i) Find the initial horizontal distance of the vehicle from the checkpoint (PV_1). [Take $\sqrt{3} = 1.732$] [1]

(ii) Find the horizontal distance of the vehicle from the checkpoint at the second observation (PV_2). [2]

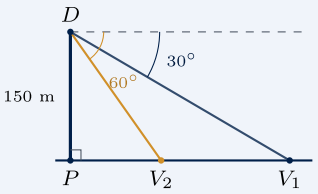
(iii) Calculate the average speed of the approaching vehicle in km/h during this 3-minute observation interval. [2]

OR

If the vehicle continues to travel at this uniform speed, how many more seconds will it take to reach the security checkpoint P? [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

(i) In right $\triangle DPV_1$ ($\angle DV_1P = 30°$):



$$PV_1 = 150 \cot 30° = 150\sqrt{3} = 150(1.732) = \boxed{259.8 \text{ m}}$$

Subpart (i): [1 Mark]

(ii) In right $\triangle DPV_2$ ($\angle DV_2P = 60°$):

$$PV_2 = \frac{150}{\sqrt{3}} = 50\sqrt{3} = 50(1.732) = \boxed{86.6 \text{ m}}$$

Subpart (ii): [2 Marks]

(iii) Main Choice: Distance $V_1V_2 = 259.8 - 86.6 = 173.2$ m in $t = 3$ min $= \frac{3}{60} = 0.05$ h. Speed $= \frac{0.1732 \text{ km}}{0.05 \text{ h}} = \boxed{3.464 \text{ km/h}}$. [2 Marks]

(iii) OR Choice: Notice $V_1V_2 = 2\,PV_2$. Uniform speed covers V_1V_2 in 3 min $= 180$ s, so remaining half distance PV_2 takes $\frac{180}{2} = \boxed{90 \text{ seconds}}$. [2 Marks]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★